

Pulse-programmed mechanical memory for zero-standby robotic matter

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A useful robotic material cell should not merely move; it should accept a brief command, enter a legible mechanical state, and hold that state without continuous power. FluxCell frames that capability as pulse-programmed mechanical memory. The present manuscript defines a corrected electropermanent-magnetic Sarrus-cell architecture in which square pole plates remain continuous, solid steel keepers translate through the working gap, and orthogonal X/Y layers create a two-axis cell from the same one-axis proof. The current contribution is deliberately bounded: a figure-grounded geometry lock, a purchase-ready proof, and the exact evidence gate needed before force, efficiency, lifetime, or reliable two-axis operation can be claimed.

1 Introduction

Soft robots and robotic materials increasingly distribute function through body geometry, material state, and embedded physical computation rather than through a single remote actuator (1–3). This direction is attractive because mechanical structure can store information, route signals, and shape motion before software has to intervene. A central unresolved question is not simply how to actuate a compliant structure, but how to make a repeated unit that changes state, stores that state physically, and tiles into a larger machine without paying continuous electrical power to remember its last command. Electropermanent magnets offer a direct route to low-standby-power state retention. An electropermanent magnet combines magnetic materials with different coercivities so that a short current pulse can switch the net external field while the magnetized state persists after current returns to zero (4–6). EPMs have been used for programmable matter, modular robotic connection, magnetic soft-robot components, and magnetorheological fluid control (4–7). In parallel, physical-interface work has shown that embedded magnets can create legible, passive haptic states in digitally fabricated mechanisms (8).

FluxCell connects those threads to a Sarrus-cell robotic-matter concept. The design does not ask a small magnet to pull a compliant mechanism across a large open gap. Instead, a fixed electropermanent magnet cartridge creates a localized magnetic circuit, while steel fork and tongue keepers slide through continuous pole-plate gap regions as the Sarrus-compatible frame changes width. The first experiment is therefore reduced to a compact overlap problem: pulse the cartridge, record whether geometry changes, check whether the state persists at zero current, reverse the pulse, and measure the limits. This is a stronger scientific object than a broad concept sketch because it can fail cleanly. Either pulse-programmed keeper overlap produces a visible held mechanical state, or the measurement identifies the dominant failure mode.

2 Results

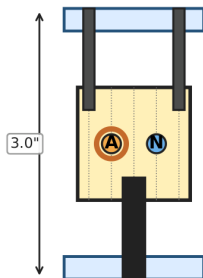
FluxCell is presently a pre-measurement design result. The result is therefore not an efficiency claim or a force-density claim. It is the conversion of a speculative robotic-matter idea into a physically ordered architecture, a locked bill of materials, and a small proof whose required traces are defined before the bench test begins.

2.1 The Y layer defines the one-axis proof

The one-axis proof is the Y layer of the final two-axis geometry (Fig. 1). It uses a lane order of fork, Alnico rod, center tongue, NdFeB rod, and fork. The full colored layer represents the physical magnetic gap region; no slots or white lanes are cut into the pole plates. This correction matters because the magnetic circuit is no longer being described as a set of separated lanes in the steel. The pole plates stay continuous, while the dark steel keepers move over the gap region.

**Y LAYER: front/back axis
EXPANDED / OFF**

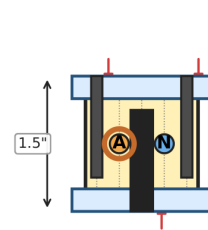
lane order: fork | A | center tongue | N | fork



full-colored layer = physical magnetic gap
(no slots/gaps are cut into the pole plates)

**Y LAYER: front/back axis
CONTRACTED / ON**

lane order: fork | A | center tongue | N | fork



full-colored layer = physical magnetic gap
(no slots/gaps are cut into the pole plates)

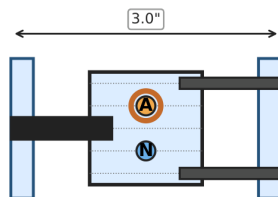
Figure 1. Y-layer one-axis proof geometry. The expanded/off state is 3.0 inches wide; the contracted/on state is 1.5 inches wide. The lane order is fork, Alnico, center tongue, NdFeB, fork. The full colored layer is the physical magnetic gap region rather than a slotted pole plate. Source: `final_y_x_layers_and_opaque_view.pdf`, created 13 May 2026 and supplied as the corrected figure source on 16 May 2026.

2.2 The X layer gives the same proof an orthogonal axis

The X layer is the same geometry rotated 90 degrees (Fig. 2). It preserves the 3.0 inch expanded/off and 1.5 inch contracted/on widths while orienting the keeper motion left-to-right. This is the simplest route from the one-axis coupon to the intended two-axis cell: prove one layer, then stack the same layer logic in the orthogonal direction. The figure also defines the packaging boundary. The Alnico and NdFeB positions remain inside the gap region; the moving fork and tongue keepers supply the visible state change.

X LAYER: left/right axis EXPANDED / OFF

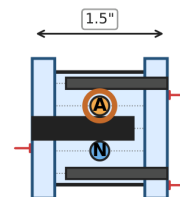
same size as Y layer, rotated 90°



full blue layer = physical magnetic gap
(no white lanes shown)

X LAYER: left/right axis CONTRACTED / ON

same size as Y layer, rotated 90°



full blue layer = physical magnetic gap
(no white lanes shown)

Figure 2. X-layer geometry. The X layer repeats the Y-layer dimensions and keeper layout, rotated 90 degrees. The full blue layer is the physical magnetic gap region, with no white lanes shown. Source: `final_y_x_layers_and_opaque_view.pdf`, created 13 May 2026 and supplied as the corrected figure source on 16 May 2026.

2.3 The assembled cell should look simple from the outside

The opaque assembled view makes the external object testable without exposing every internal lane (Fig. 3). The top steel pole plate hides the Alnico and NdFeB rods and the internal magnetic gaps. The observer should see the top plate, carriers, and protruding steel keepers. This is important for the physical proof because the visible object should not require the reader to infer hidden geometry from a complicated exploded view. A successful bench specimen should make the same external state change legible on video while the current, temperature, and width traces provide the evidence.

OPAQUE ASSEMBLED VIEW — what it would look like externally

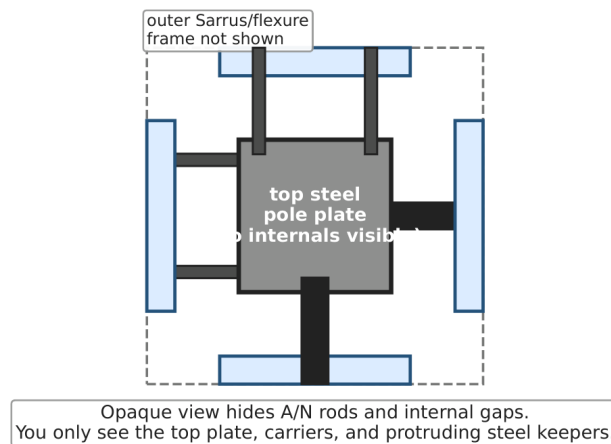


Figure 3. Opaque assembled view. The external view hides A/N rods and internal gaps; only the top plate, carriers, and protruding steel keepers remain visible. The outer Sarrus/fluxure frame is not shown. Source: `final_y_x_layers_and_opaque_view.pdf`, created 13 May 2026 and supplied as the corrected figure source on 16 May 2026.

2.4 Construction schematics expose the magnetic state

The construction views are intentionally plain schematics, not a rendered final product. The state view opens the cell enough to show the X and Y layers together, the Alnico and NdFeB rods in each layer, and the expanded/off versus contracted/on keeper positions (Fig. 4). The fork prongs terminate flush at the carrier-bar face rather than passing through the carrier. The exploded stack view then separates the top, shared center, and bottom pole plates enough to show how the two orthogonal magnetic layers occupy the same cell footprint (Fig. 5). No Sarrus/flexure shell, wire routing, fasteners, or decorative housing is added.

Two-layer magnetic layout: expanded and contracted

Both panels show the X and Y layers together. Orange rods are Alnico; blue rods are NdFeB.

● Alnico ● NdFeB

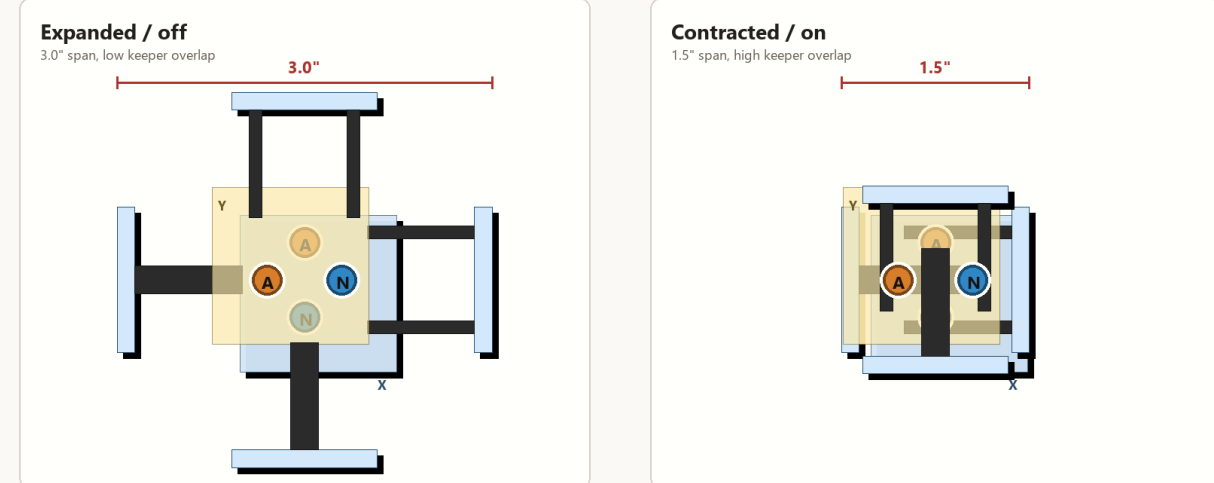


Figure 4. Opened two-layer magnetic state schematic. Both panels show the X and Y layers together. Orange rods mark Alnico; blue rods mark NdFeB. The Y-layer A/N pair is left-right, and the rotated X-layer A/N pair is top-bottom. The muted top-bottom pair is the lower X layer seen through the translucent Y layer. The expanded/off state shows the 3.0 inch span and low keeper overlap; the contracted/on state shows the 1.5 inch span and deeper overlap. The fork prongs end flush at the carrier-bar face.

Exploded top-down construction view

Top-down view. The single center tongue is between Alnico/NdFeB; the fork prongs are outside.

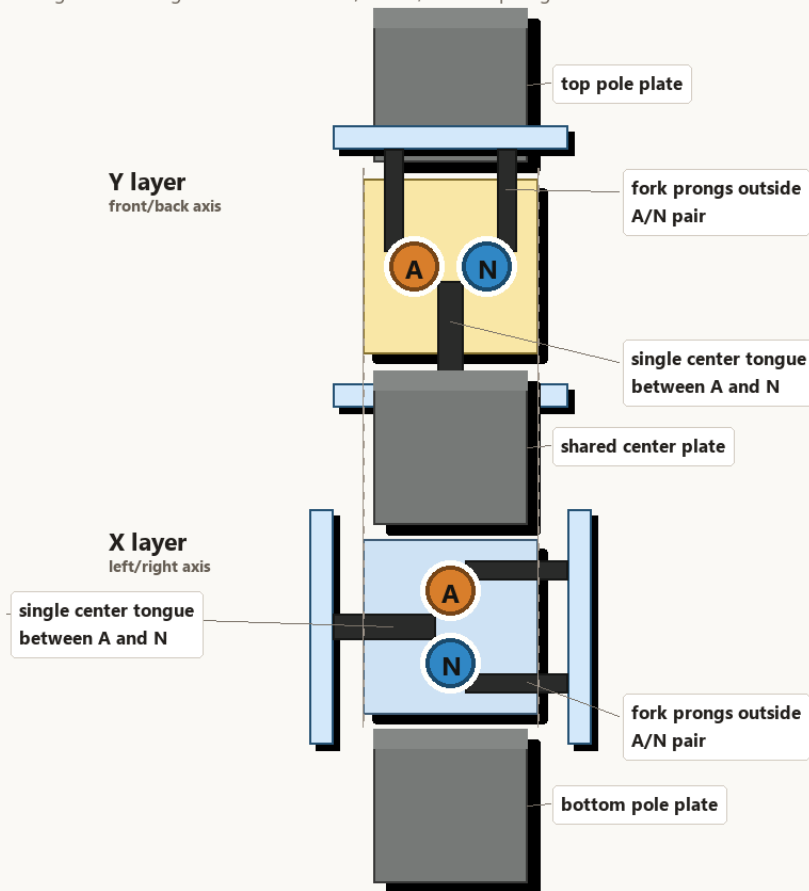


Figure 5. Exploded top-down construction schematic. The top, shared center, and bottom pole plates are separated to reveal the two orthogonal magnetic layers. In each layer, the single center tongue runs between the Alnico and NdFeB rods, while the fork prongs run outside the rod pair. The view adds no shell, wiring, or decorative housing.

2.5 The dimensional lock creates a falsifiable mechanical-memory proof

The current dimensional correction uses 1.00 inch keeper inserts rather than 1-1/8 inch inserts. With 1.25 inch square pole plates, the one-axis proof targets a 3.0 inch expanded/off width and a 1.5 inch contracted/on width, corresponding to a 0.75 inch side stroke on each side. The off-state overlap is approximately 0.125 inch and the on-state overlap is approximately 0.875 inch. These numbers are design targets, not measured performance. Their value is that they make the first physical test falsifiable: a synchronized record can show width, pulse polarity, current, hold at zero drive, release after reverse pulse, and coil temperature.

2.6 The publishable evidence gate is a synchronized trace

The manuscript becomes an actuator paper only if the one-axis proof produces a synchronized record rather than a single success photograph. The minimum publishable trace is deliberately small (Table 1). It has to show the command, the electrical cost of the command, the mechanical response, and whether the state remains after drive current returns to zero. This makes the evidence portable: the same trace can support a robotics claim, a mechanical-memory claim, and a later four-cell surface demonstration.

2.7 The build lock is small enough to purchase and test

The total before shipping and tax is approximately \$456.65 (9–12). This number is not a claim about manufacturability at scale. It is a friction claim: the proof has a specific bill of materials and no longer depends on an undefined part-search step.

Table 1. Evidence gate for converting the present design manuscript into an experimental actuator manuscript.

Signal	Required record	Current state
Stroke	Width before pulse, width after pulse, and held width at zero current	pending measurement
Switching cost	Pulse polarity, duration, current limit, and supply voltage	pending measurement
Hold	Zero-current interval with no visible return drift	pending measurement
Release	Opposite pulse followed by measurable release or reverse motion	pending measurement
Thermal margin	Coil temperature note after repeated pulses	pending measurement
Failure mode	If motion fails, dominant limit: friction, weak flux, twist, heat, or residual sticking	pending measurement

Table 2. Locked purchase logic for one proof plus four complete two-axis cells. Costs are purchase-planning values before shipping and tax, not experimental performance data.

Part	Qty	Locked specification	Approx total
Alnico 5 rods	12	0.250 inch diameter by 0.625 inch long, axial	\$23.88
NdFeB rods	12	K&J D4A, 1/4 inch diameter by 5/8 inch long, N42 axial	~\$15.00
Square pole plates	16	1018 steel, 0.25 inch thick by 1.25 inch wide, cut length 1.25 inch	~\$56.00
Fork prong inserts	24	A36 flat bar, 1/8 inch by 1/2 inch, cut length 1.00 inch	\$167.28
Center tongue inserts	12	1018 flat bar, 1/4 inch by 1/2 inch, cut length 1.00 inch	\$87.72
Wire, tape, supply, leads, pigtails, heat shrink	set	30 AWG magnet wire, Kapton, 0–30 V / 0–10 A supply, leads, silicone wire, strain relief	~\$106.75

3 Discussion

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The strongest version of FluxCell is not a generic magnetic actuator paper. It is a paper about pulse-programmed mechanical memory. The state variable is not merely magnetic polarity; it is a visible mechanical configuration coupled to magnetic overlap and held after the electrical pulse ends. If the one-axis proof records clean switching, zero-current hold, reverse-pulse release, bounded heating, and repeated cycles, the work can move from design protocol to actuator evidence.

The source boundary matters because overclaiming would make the project weaker, not stronger. This manuscript does not claim force density, efficiency, safety, lifetime, or reliable two-axis behavior. It claims that the current design has become concrete enough to test and that the proof has been reduced to a small set of measurements. That reduction is the paper's present contribution. It makes the next experiment difficult to hand-wave: either the geometry moves and holds under a measured pulse, or the data identify the dominant failure mode.

For R&D in physical experiences, the compelling artifact would be a small surface that changes contour, holds while electrically quiet, and returns on command. That future demonstrator would be human-readable because the motion, latch, and reset are visible. This is the portfolio-facing reason the project matters for an Imagineering-style R&D path: it combines physical computing, visible mechanism design, soft-robotic structure, rapid fabrication, and evidence-rich iteration in one inspectable artifact. The current paper does not claim that demonstrator exists, nor does it imply any hiring outcome or institutional relationship. It defines the cell-level experiment that must pass before that story is credible.

4 Materials and Methods

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4.1 One-axis proof geometry

The proof uses a fixed center EPM cartridge with two 1.25 inch square pole plates made from 0.25 inch by 1.25 inch 1018 steel rectangle bar cut to 1.25 inch length. The magnetic elements are one 0.250 inch diameter by 0.625 inch long Alnico 5 rod and one 1/4 inch diameter by 5/8 inch long NdFeB rod. The moving keepers are two A36 fork prong inserts, each 1/8 inch by 1/2 inch by 1.00 inch, and one 1018 center tongue insert, 1/4 inch by 1/2 inch by 1.00 inch. The Y/X/opaque figures in this draft are cropped from the supplied source PDF without changing figure content (13). The 3D construction schematics are generated from these dimensions and include only parts described in the paper.

4.2 Electrical pulse protocol

The Alnico coil is insulated with Kapton, wound with 30 AWG magnet wire, and connected to an H-bridge driver from a current-limited bench supply. The first test uses short bidirectional pulses separated by zero-drive intervals. For each pulse, the record should include polarity, pulse duration, current limit, observed width, hold state after zero drive, release state after reverse pulse, and a coil temperature note.

4.3 Acceptance boundary

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The first proof succeeds only if the one-axis assembly visibly changes width, holds a state after current returns to zero, releases or reverses under the opposite pulse, and stays within a temperature range safe enough for continued bench testing. A partial result remains useful if it identifies the dominant failure mode: weak magnetic force, high friction, carrier twist, thermal limit, or poor release margin.

5 Acknowledgements

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This draft was assembled from the FluxCell design log, the corrected Y/X/opaque geometry source PDF, and the current locked purchase list.

6 Funding

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No external funding is claimed in this draft.

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7 Author contributions

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Alan N. Pham conceived the project direction, maintained the design log, selected the current purchase lock, and defined the proof target. Codex assisted with manuscript formatting, page integration, figure placement, and verification workflow.

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8 Competing interests

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The author declares no competing interests for this draft.

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9 Data availability

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The current data are design records, the supplied corrected geometry PDF, cropped figure exports, generated 3D construction schematics, supplier links, and purchase specifications. Measured actuation data are pending the one-axis proof.

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10 Code availability

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The current project page, generated proof bundle, manuscript source, and 3D schematic rendering script are served through the FluxCell site. Arduino pulse code is included in the generated proof bundle.

11 Additional information

This is a draft manuscript and design protocol, not a completed experimental report. The claims should be updated only after the one-axis proof records measured current, width, temperature, hold, release, and repeatability.

12 References

1. D. Rus, M. T. Tolley, Design, fabrication and control of soft robots. *Nature* **521**, 467–475 (2015).
2. M. Wehner *et al.*, An integrated design and fabrication strategy for entirely soft, autonomous robots. *Nature* **536**, 451–455 (2016).
3. J. Byun, A. Pal, J. Ko, *et al.*, Integrated mechanical computing for autonomous soft machines. *Nature Communications* **15**, 2933 (2024).
4. A. N. Knaian, PhD thesis, Massachusetts Institute of Technology, 2010.
5. W. R. Johnson, R. Kramer-Bottiglio, presented at the 2024 IEEE 7th International Conference on Soft Robotics (RoboSoft).
6. L. Zhu, D. El Baz, A programmable actuator for combined motion and connection and its application to modular robot. *Mechatronics* **58**, 9–19 (2019).
7. V. Ntella, A. Naik, L. Kinnicutt, J. Li, T. Ranzani, Highly Efficient Miniaturized Magnetorheological Valves Using Electropermanent Magnets. *IEEE Robotics and Automation Letters* **8**, 1535–1542 (2023).
8. C. Zheng, J. Kim, D. Leithinger, M. D. Gross, E. Y.-L. Do, presented at the Proceedings of the 13th International Conference on Tangible, Embedded, and Embodied Interaction, pp. 325–334.
9. A. N. Pham, *FluxCell design log: square pole plates, solid inserts, one-axis proof, and four-cell purchase lock*, Project source, uploaded design figures, and purchase lock recorded in FluxCell, May 2026, 2026.

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10. OnlineMetals, *1018 cold-finish steel rectangle bar, 0.25 inch by 1.25 inch* (2026; <https://www.onlinemetals.com/en/buy/carbon-steel/0-25-x-1-25-carbon-steel-rectangle-bar-1018-cold-finish/pid/7466>). 169
170
171
11. All Metals Inc., *A36 hot rolled flat bar, 1/8 inch by 1/2 inch* (2026; <https://allmetalsinc.com/products/steel-hot-rolled-flat-bar-1-8-x-1-2-grade-a36>). 172
173
12. All Metals Inc., *1018 cold rolled flat bar, 1/4 inch by 1/2 inch* (2026; <https://allmetalsinc.com/products/steel-cold-rolled-flat-bar-1-4-x-1-2-grade-1018>). 174
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13. A. N. Pham, *Final simplified views: Y layer, X layer, and opaque assembled view*, FluxCell 176
corrected geometry source PDF, created 13 May 2026 and supplied 16 May 2026, 2026. 177